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○ rishitkamboj@Rishits-MacBook-Air-2 proj1 % python3 src/app.py
Loading corpus and building indexes ...
Ready: 100 documents, 124 terms indexed.
```

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Clothing Search Engine (100 products)
=====
1) Free-text ranked search (VSM, lnc.ltc cosine)
2) Exact phrase search (positional index)
3) Proximity search WITHIN/k (positional index)
4) Smart search [NOVELTY] (VSM + phrase + proximity)
0) Quit
-----
Select mode (0-4): █
```

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-----
Select mode (0-3): 1
Enter free-text query: cotton shirt
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```
query terms after preprocessing: ['cotton', 'shirt']
Top 10 results (lnc.ltc cosine):
```

rank	docID	score	category / title
1	D001	0.2731	T-Shirt Men's Cotton Crew Neck T-Shirt - Black
2	D041	0.2731	T-Shirt Men's Cotton Crew Neck T-Shirt - Maroon
3	D061	0.2731	T-Shirt Men's Cotton Crew Neck T-Shirt - White
4	D081	0.2703	T-Shirt Men's Cotton Crew Neck T-Shirt - Mustard
5	D022	0.2701	Shirt Men's Checked Cotton Shirt - White
6	D082	0.2701	Shirt Men's Checked Cotton Shirt - Blue
7	D042	0.2683	Shirt Men's Checked Cotton Shirt - Maroon
8	D002	0.2654	Shirt Men's Checked Cotton Shirt - Black
9	D062	0.2654	Shirt Men's Checked Cotton Shirt - Olive
10	D021	0.2643	T-Shirt Men's Cotton Crew Neck T-Shirt - Sky Blue

```
-----
Select mode (0-3): 2
Enter exact phrase (e.g. cotton shirt): cotton shirt
```

```
phrase terms after preprocessing: ['cotton', 'shirt']
5 document(s) contain the phrase (showing start positions as evidence):
```

D002	start_pos=[3, 9]	Shirt	Men's Checked Cotton Shirt - Black
D022	start_pos=[3, 9]	Shirt	Men's Checked Cotton Shirt - White
D042	start_pos=[3, 9]	Shirt	Men's Checked Cotton Shirt - Maroon
D062	start_pos=[3, 9]	Shirt	Men's Checked Cotton Shirt - Olive
D082	start_pos=[3, 9]	Shirt	Men's Checked Cotton Shirt - Blue

Mode 1 cotton shirt (t-shirts rank high) vs Mode 2 cotton shirt (only 5 checked shirts)

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=====
Select mode (0-3): 2
Enter exact phrase (e.g. cotton shirt): high waist

phrase terms after preprocessing: ['high', 'waist']
10 document(s) contain the phrase (showing start positions as evidence):

D009 start_pos=[2, 10, 32] Leggings | Women's High Waist Stretch Leggings - Olive Green
D019 start_pos=[28] Leggings | Women's Printed Active Leggings - Black
D029 start_pos=[2, 9, 30] Leggings | Women's High Waist Stretch Leggings - Grey
D039 start_pos=[30] Leggings | Women's Printed Active Leggings - Olive Green
D049 start_pos=[2, 9, 30] Leggings | Women's High Waist Stretch Leggings - Black
D059 start_pos=[28] Leggings | Women's Printed Active Leggings - Grey
D069 start_pos=[2, 10, 32] Leggings | Women's High Waist Stretch Leggings - Olive Green
D079 start_pos=[28] Leggings | Women's Printed Active Leggings - Black
D089 start_pos=[2, 9, 30] Leggings | Women's High Waist Stretch Leggings - Grey
D099 start_pos=[30] Leggings | Women's Printed Active Leggings - Olive Green
=====

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=====
Select mode (0-3): 3
Enter proximity query (e.g. cotton WITHIN/3 shirt): stretch WITHIN/5 denim

'stretch' followed by 'denim' within 5 positions (ordered):
7 document(s) match (showing matching (pos_stretch, pos_denim) pairs as evidence):

D003 pairs=[[17, 18]] Jeans | Men's Regular Fit Denim Jeans - Grey
D013 pairs=[[16, 17]] Jeans | Men's Slim Fit Stretch Jeans - Grey
D033 pairs=[[17, 18]] Jeans | Men's Slim Fit Stretch Jeans - Grey
D043 pairs=[[16, 17]] Jeans | Men's Regular Fit Denim Jeans - Grey
D063 pairs=[[17, 18]] Jeans | Men's Regular Fit Denim Jeans - Grey
D073 pairs=[[16, 17]] Jeans | Men's Slim Fit Stretch Jeans - Grey
D093 pairs=[[17, 18]] Jeans | Men's Slim Fit Stretch Jeans - Grey
=====

```

Mode 1 vs Mode 2 for “Festive wear”:

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=====
Select mode (0-3): 1
Enter free-text query: festive wear

query terms after preprocessing: ['festiv', 'wear']
No rankable results - every query term is out-of-vocabulary or
occurs in all documents (idf 0). Try mode 2/3 for such phrases.
=====

```

Clothing Search Engine (100 products)

- ```

=====
1) Free-text ranked search (VSM, lnc.ltc cosine)
2) Exact phrase search (positional index)
3) Proximity search WITHIN/k (positional index)
0) Quit
=====

```

```

Select mode (0-3): 2
Enter exact phrase (e.g. cotton shirt): festive wear

```

```

phrase terms after preprocessing: ['festiv', 'wear']
10 document(s) contain the phrase (showing start positions as evidence):

```

```

D005 start_pos=[27] Saree | Women's Printed Daily Wear Saree - Blue
D015 start_pos=[25] Saree | Women's Cotton Handloom Saree - Red
D025 start_pos=[27] Saree | Women's Printed Daily Wear Saree - Maroon
D035 start_pos=[25] Saree | Women's Cotton Handloom Saree - Yellow
D045 start_pos=[27] Saree | Women's Printed Daily Wear Saree - Purple
D055 start_pos=[25] Saree | Women's Cotton Handloom Saree - Pink
D065 start_pos=[27] Saree | Women's Printed Daily Wear Saree - Green
D075 start_pos=[25] Saree | Women's Cotton Handloom Saree - Blue
D085 start_pos=[27] Saree | Women's Printed Daily Wear Saree - Red
D095 start_pos=[25] Saree | Women's Cotton Handloom Saree - Maroon

```

## MODE 4: (NOVELTY)

```

Select mode (0-4): 4
Enter query for smart search: cotton shirt

query terms after preprocessing: ['cotton', 'shirt']
Smart-ranked results (net = cosine + phrase-bonus + proximity):

```

| rank | docID | net    | cosine | signal | category / title                                    |
|------|-------|--------|--------|--------|-----------------------------------------------------|
| 1    | D022  | 0.9201 | 0.2701 | phrase | Shirt   Men's Checked Cotton Shirt - White          |
| 2    | D082  | 0.9201 | 0.2701 | phrase | Shirt   Men's Checked Cotton Shirt - Blue           |
| 3    | D042  | 0.9183 | 0.2683 | phrase | Shirt   Men's Checked Cotton Shirt - Maroon         |
| 4    | D002  | 0.9154 | 0.2654 | phrase | Shirt   Men's Checked Cotton Shirt - Black          |
| 5    | D062  | 0.9154 | 0.2654 | phrase | Shirt   Men's Checked Cotton Shirt - Olive          |
| 6    | D001  | 0.3481 | 0.2731 | win=4  | T-Shirt   Men's Cotton Crew Neck T-Shirt - Black    |
| 7    | D041  | 0.3481 | 0.2731 | win=4  | T-Shirt   Men's Cotton Crew Neck T-Shirt - Maroon   |
| 8    | D061  | 0.3481 | 0.2731 | win=4  | T-Shirt   Men's Cotton Crew Neck T-Shirt - White    |
| 9    | D081  | 0.3453 | 0.2703 | win=4  | T-Shirt   Men's Cotton Crew Neck T-Shirt - Mustard  |
| 10   | D021  | 0.3393 | 0.2643 | win=4  | T-Shirt   Men's Cotton Crew Neck T-Shirt - Sky Blue |

```

Select mode (0-4): 4
Enter query for smart search: festive wear

query terms after preprocessing: ['festiv', 'wear']
Smart-ranked results (net = cosine + phrase-bonus + proximity):

```

| rank | docID | net    | cosine | signal | category / title                                  |
|------|-------|--------|--------|--------|---------------------------------------------------|
| 1    | D005  | 0.6500 | 0.0000 | phrase | Saree   Women's Printed Daily Wear Saree - Blue   |
| 2    | D015  | 0.6500 | 0.0000 | phrase | Saree   Women's Cotton Handloom Saree - Red       |
| 3    | D025  | 0.6500 | 0.0000 | phrase | Saree   Women's Printed Daily Wear Saree - Maroon |
| 4    | D035  | 0.6500 | 0.0000 | phrase | Saree   Women's Cotton Handloom Saree - Yellow    |
| 5    | D045  | 0.6500 | 0.0000 | phrase | Saree   Women's Printed Daily Wear Saree - Purple |
| 6    | D055  | 0.6500 | 0.0000 | phrase | Saree   Women's Cotton Handloom Saree - Pink      |
| 7    | D065  | 0.6500 | 0.0000 | phrase | Saree   Women's Printed Daily Wear Saree - Green  |
| 8    | D075  | 0.6500 | 0.0000 | phrase | Saree   Women's Cotton Handloom Saree - Blue      |
| 9    | D085  | 0.6500 | 0.0000 | phrase | Saree   Women's Printed Daily Wear Saree - Red    |
| 10   | D095  | 0.6500 | 0.0000 | phrase | Saree   Women's Cotton Handloom Saree - Maroon    |

## EVALUATION HARNESS (NOVELTY):

```

● rishitkamboj@Rishits-MacBook-Air-2 proj1 % python3 scripts/evaluate.py
EVALUATION - Precision@k : plain VSM vs Smart Search (novelty)
Relevance = retrieved doc's CATEGORY matches the query's garment type.

```

| query               | relevant   | VSM P@5 | Smart P@5 | VSM P@10 | Smart P@10 |
|---------------------|------------|---------|-----------|----------|------------|
| denim jeans         | Jeans      | 1.00    | 1.00      | 1.00     | 1.00       |
| printed saree       | Saree      | 1.00    | 1.00      | 1.00     | 1.00       |
| midi dress          | Dress      | 1.00    | 1.00      | 1.00     | 1.00       |
| fleece hoodie       | Hoodie     | 1.00    | 1.00      | 1.00     | 1.00       |
| winter jacket       | Jacket     | 1.00    | 1.00      | 1.00     | 1.00       |
| high waist leggings | Leggings   | 1.00    | 1.00      | 1.00     | 1.00       |
| graphic t shirt     | T-Shirt    | 1.00    | 1.00      | 0.50     | 0.50       |
| cotton shirt        | Shirt      | 0.20    | 1.00      | 0.50     | 0.50       |
| regular fit kurta   | Kurta      | 1.00    | 1.00      | 1.00     | 1.00       |
| fleece sweatshirt   | Sweatshirt | 1.00    | 1.00      | 1.00     | 1.00       |
| MEAN                |            | 0.92    | 1.00      | 0.90     | 0.90       |

```

Mean P@5 : VSM 0.920 -> Smart 1.000 (+0.080)
Mean P@10: VSM 0.900 -> Smart 0.900 (+0.000)

Smart Search never scores lower than VSM, and improves precision on
ambiguous queries (e.g. 'cotton shirt', where VSM ranks T-shirts high)
by promoting exact-phrase / close-proximity matches.

```

## Mandatory Tests:

```
● rishitkamboj@Rishits-MacBook-Air-2 proj1 % python3 tests/test_positional.py
Loaded 100 docs, 124 terms.

[1] Exact phrase queries
'cotton shirt': 5 docs -> ['D002', 'D022', 'D042', 'D062', 'D082']
'stretch denim': 7 docs -> ['D003', 'D013', 'D033', 'D043', 'D063', 'D073', 'D093']
'festive wear': 10 docs -> ['D005', 'D015', 'D025', 'D035', 'D045', 'D055', 'D065', 'D075', 'D085', 'D095']
'winter wear': 20 docs -> ['D008', 'D010', 'D018', 'D020', 'D028', 'D030', 'D038', 'D040', 'D048', 'D050', 'D058', 'D060', 'D068',
'D070', 'D078', 'D080', 'D088', 'D090', 'D098', 'D100']
'high waist': 10 docs -> ['D009', 'D019', 'D029', 'D039', 'D049', 'D059', 'D069', 'D079', 'D089', 'D099']

[2] Ordered proximity queries (different k)
cotton WITHIN/2 shirt: 5 docs -> ['D002', 'D022', 'D042', 'D062', 'D082']
stretch WITHIN/4 denim: 7 docs -> ['D003', 'D013', 'D033', 'D043', 'D063', 'D073', 'D093']
winter WITHIN/3 wear: 20 docs -> ['D008', 'D010', 'D018', 'D020', 'D028', 'D030', 'D038', 'D040', 'D048', 'D050', 'D058', 'D060',
'D068', 'D070', 'D078', 'D080', 'D088', 'D090', 'D098', 'D100']

[3] Phrase-vs-proximity relationships
phrase(cotton shirt)=5 <= WITHIN/3=10; WITHIN/2=5 <= WITHIN/4=15

[4] Empty-result phrase
'zip closure': 0 docs

[5] Out-of-vocabulary term
'silk' absent -> phrase and proximity both empty (as expected)

All positional/phrase/proximity checks passed (298 assertions).
○ rishitkamboj@Rishits-MacBook-Air-2 proj1 %
```

```
● rishitkamboj@Rishits-MacBook-Air-2 proj1 % python3 tests/test_vsm.py
PASS 0.8 lecture example
PASS 10 free-text queries
PASS out-of-vocabulary -> []
PASS all idf 0 -> []

All 4 tests passed.
○ rishitkamboj@Rishits-MacBook-Air-2 proj1 %
```

```
● rishitkamboj@Rishits-MacBook-Air-2 proj1 % python3 tests/test_smart_search.py
Loaded 100 docs.

[1] smallest_window
[2] phrase matches promoted above non-phrase (cotton shirt)
[3] 'festive wear': VSM empty, Smart Search returns sarees
[4] net >= cosine and sorted by net
[5] single-term query == plain VSM ordering

All Smart Search checks passed (21 assertions).
○ rishitkamboj@Rishits-MacBook-Air-2 proj1 %
```